

Unwrapped Hub Surface — Cylindrical Mapping

$$\theta = \arctan\left(\frac{y}{x}\right), \quad s = r \cdot \theta, \quad t = z$$

- **Isometric mapping:** distortion-free, preserves triangulation
- **Domain becomes rectangular** with a blade-shaped hole
- **Periodic θ -boundaries** (pitch = $\pi/4$ for 8 blades)

Code: `scripts/stage1/unwrap_surface.py:8-10`